DERIVED COMPLETION OF A COMPLEX WITH FINITE COHOMOLOGY COMPUTES THE ADIC COMPLETIONS OF ITS COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span writes 'H^n(K) a finite A-module for all n', leaving the verb implicit; that wording is delivered verbatim and not corrected. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- A bounded-above complex over a Noetherian ring with finite cohomology is pseudo-coherent and has a finite free model. invoked at proof lines 27275-27278. Produces the finite free complex P^bullet representing each truncation.
- The naive completion of a Noetherian ring agrees with the derived completion, and R lim of a system of modules is computed termwise. invoked at proof lines 27280-27283. Identifies the derived completion of the truncation with the termwise limit of P^bullet/I^n.
- I-adic completion is exact on finite modules, and derived completion has finite cohomological dimension. invoked at proof lines 27284-27288. Gives the statement for each truncation and then passes to K.

Source reference. Original lemma 00MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 817–817. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 16517: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16517–16517. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 23574: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23574–23574. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 26508: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26508–26508. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original proposition 27218: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27218–27218. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a Noetherian ring and $I \subset A$ an ideal. Let K be an object of $D(A)$ such that $H^n(K)$ a finite A-module for all $n \in \mathbf{Z}$. Then the cohomology modules $H^n(K^\wedge)$ of the derived completion are the I-adic completions of the cohomology modules $H^n(K)$.*

Proof. The complex $\tau_{\leq m}K$ is pseudo-coherent for all m by Lemma 16517. Thus $\tau_{\leq m}K$ is represented by a bounded above complex $P^\bullet$ of finite free A-modules. Then $\tau_{\leq m}K \otimes^{\mathbf{L}}_A A/I^n = P^\bullet/I^nP^\bullet$. Hence $(\tau_{\leq m}K)^\wedge = R\lim P^\bullet/I^nP^\bullet$ (Proposition 27218) and since the $R\lim$ is just given by termwise lim (Lemma 23574) and since I-adic completion is an exact functor on finite A-modules (Algebra, Lemma 00MB) we conclude the result holds for $\tau_{\leq m}K$. Hence the result holds for K as derived completion has finite cohomological dimension, see Lemma 26508. □